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MILITARY SPECIFICATION

POWER SUPPLY, GUIDED MISSILE AND GUIDED MISSILE FIRE CONTROL SYSTEM SUPPORT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rectifier power supplies for naval submarine use in energizing missile fire control systems and missile components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-C-2212	- Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-S-3950 - Switches, Toggle, Environmentally Sealed, General Specification for.
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Housing (Shipboard Use).
MIL-S-17773 - Switching Unit, Power Transfer (Bus Transfer) (Naval Shipboard Use).
MIL-M-28787 - Modules, Standard Electronic General Specification for.

STANDARDS

MILITARY

MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric).

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0967-LP-312-8010 - Electronic Equipment, Shipboard, Maintainability Design Criteria Handbook for Designers of.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The power supply shall be in accordance with class R of MIL-P-15736 and as specified herein.

3.1.1 Summary of requirements. The power supply shall operate as specified herein for any combination of resistive, inductive, capacitative or motor load within the specific load rating. Requirements of table I shall apply, and except where noted, all parameters listed refer to output characteristics.

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TABLE I. Summary of requirements.

Parameter	Requirement
A.c. V (input) Frequency (input) Phase (input)	As specified in DOD-STD-1399, Section 300, type I power, 440 V, 3 phase
D.c. voltage	Adjustable 26 to 32 V (see 3.2.2.1)
D.c. current	480 A rated full load, continuous (see 3.2.2.2) 500 A intermittent 2 second duration
Duty	Continuous (see 3.2.2.2)
Enclosure	Dripproof protected (see 3.3.1)
Type of load	Resistive, inductive, capacity and motor 0.9 pf leading or lagging as minimum
Ambient temperature range	23°C (room temperature) to 50°C
Ventilation	Natural draft
Mounting	Deck-suitable for center of gravity mounting for use on sound isolation mounts (see 3.3.1)
Meters	Voltmeter 0-50 V, Ammeter 0-500 A (digital instruments are described in 3.3.2.1)
Size and configuration	See 3.3.1
Weight	1250 pounds, maximum
Shock	Grade A, deck mounted, class I, medium weight, type A

3.2 Performance requirements.

3.2.1 Alternating current input. The power supply shall operate from type I power as described by DOD-STD-1399, Section 300, except as otherwise specified herein. Attention of designers is directed to the transient characteristics of the input voltage described by DOD-STD-1399, Section 300.

3.2.1.1 Input power interruption. The power supply shall withstand input power interruption of duration ranging from 20 milliseconds (ms) to 50 ms resulting from transfer from one power source to an unsynchronized alternate power source. Equipment shall incur no damage to any component when the alternating current (a.c.) line input voltage to the equipment is interrupted

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over the intervals specified above, and then restored. Upon reapplication of the 440 volt (V) a.c. input line, the direct current (d.c.) output voltage of the power supply is permitted to overshoot above the regulated set point of d.c. voltage, provided that output does not exceed 35 V d.c. (see 3.2.2.1), 200 ms. Output voltage shall recover to the regulated set point within 200 ms following decay of transients. The power supply shall be so designed that the blowing of fuses or tripping of input contactor, solely as a result of the power transfer, does not occur (see 3.3.2(d)).

3.2.1.2 Harmonic distortion of input current. With the exception of 11th and 13th harmonics of power frequency, the magnitude of any individual line input harmonic currents, from the second through the 32nd, expressed as a percentage of the full load fundamental root mean square (rms) line current, shall not exceed 3 percent for any input current between no load and rated full load value. Similarly, no individual harmonic line current from the 33rd harmonic through 20 kilohertz (kHz) shall exceed a magnitude in percent equal to $1700/n^2$, where "n" is the harmonic multiple number. Harmonic currents shall be measured as specified in 4.2.2.

3.2.1.2.1 For any input current drawn from the 440 V line when the power supply is operated at output currents between no load and 480 amperes (A), harmonic line currents of order 11 and 13, expressed as a percentage of full load fundamental frequency rms line current, shall not exceed the percentages specified below:

(a) 6.5 percent for 11th harmonic
(b) 4.5 percent for 13th harmonic

3.2.2 D.c. output.

3.2.2.1 Output voltage. Output voltage shall be as follows:

(a) Adjustment range. Output voltage shall be adjustable from 26 to 32 V under all specified conditions of input voltage and output load.

(b) Steady-state voltage regulation. Output voltage shall be automatically regulated to within plus or minus 1 percent from no load to 480 A load and with specified steady state variation of input voltage. Power supply shall regulate voltage at a point remote from its terminals. Voltage drop between the remote point and the power supply may be as much as 3.2 V at 480 A load. However, the power supply need not furnish more than 32 V at its terminals.

(c) Transient voltage regulation load changes. Output voltage at terminals of power supply shall not drop below 24.5 V nor rise above 35 V with step load changes as follows:

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From Amperes	To Amperes
0	60
60	120
120	240
240	480
480	240
240	120
120	60
60	0

These limits under partial load changes shall apply when output voltage is initially adjusted to 28 V, and input voltage value is within specified range. For step increase of load from 240 A to 480 A, and step decrease of load from 480 A to 240 A, the output voltage of the power supply shall recover to, and remain within, the steady state regulation band of plus or minus 1 percent about 28 V d.c. within a time interval not to exceed 200 ms following step change of load. No sustained oscillations in power supply output voltage shall develop.

(d) Transient voltage regulation - input voltage transients. When the power supply is initially adjusted to 28 V at the output terminals and is supplying any load up to 300 A, the output voltage shall not drop below 24.5 V nor rise above 35 V when input voltage dips or rises 18 percent of rated value (440 V). The input voltage shall return to, and remain within, 5 percent of rated value within 2 seconds after initial disturbance. During recovery, the input voltage may overshoot in a direction opposite to that of the original disturbance. Approximately half of the voltage excursion shall occur within the first 1/120 second, and the maximum of value of the excursion shall occur between 1/10 and 3/10 second after initiation of the disturbance.

(e) Overvoltage protection. Design of the power supply shall be such that failure of circuit components or connections between components, occurring singly or in combination, shall not cause output voltage to exceed 35 V d.c. at the power supply.

3.2.2.2 Output current. Power supply shall furnish 480 A output current as continuous rated full load on the power supply.

3.2.2.3 Ripple. The ripple content of the output voltage shall not exceed 0.5 percent of the preset d.c. voltage as an rms measurement. This ripple requirement shall apply to any load ranging from no load to 480 A resistive load with balanced input a.c. line voltage.

3.2.2.4 Overcurrent requirement.

3.2.2.4.1 Overcurrent operation. When operating into a load fault through line resistance which produces output voltage drop of 3.2 V at 480 A (see 3.2.2.1(b)), the power supply shall furnish sufficient output fault current to force the tripping of an AQB-A250 circuit breaker conforming to MIL-C-17361

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and having a type 225N trip element rated at 225 A. The power supply shall have the capability to sustain an output fault current of magnitude equal to 320 percent of 225 A for an interval between 1 second and 4 seconds. Current limited operation of the power supply during overload is acceptable, providing that current limiting does not interfere with tripping of the AQB-A250 circuit breaker. Short circuit and overload faults on the output bus shall be cleared without damage to any component of the power supply and without interruption of power supply operation by blowing of fuses or overload trip of the input line contactor. After the output load fault has been cleared by tripping of the AQB-A250 circuit breaker, the d.c. power supply shall revert to voltage regulated operations in accordance with 3.2.2.1.

3.2.2.4.2 Overcurrent protection. The power supply shall be protected against overload and short circuit on output terminals. Protection shall be provided by fuses and input overload relays applied in accordance with MIL-P-15736. The use of electronic shutdown circuits which automatically disable the forward conduction of power semiconductors during output short circuit or overload is prohibited. Crowbar circuits are also prohibited as a method to protect the power supply against overload or short circuit on output terminals. Manual restart of the power supply is acceptable, in the event that input power has been interrupted by overload trip of the input magnetic contactor (see 3.3.2(a)) under conditions of overload on the output terminals of the power supply.

3.3 Construction requirements.

3.3.1 Physical size and configuration. Physical size and configuration shall be as follows:

(a) Power supply shall be packaged so that it can be handled through both a 25-inch diameter hatch and a watertight door opening 38 by 20 inches in size (10-inch radius corners) with minimum disassembly. It is permissible to construct the power supply such that it consists of up to three major subassemblies each of which can pass through the specified hatch and door. If the power supply is designed in subassemblies as described above, terminals and wiring for interconnection of these subassemblies shall be furnished.

(b) When assembled (see (a)), power supply dimensions shall not exceed the following:

Width	-	36-1/2 inches
Depth	-	18 inches
Height	-	60 inches

The depth dimension excludes extension due to horizontal surfaces at mounting attachment points for sound isolation mounting of the power supply (see (d)).

(c) Power supply shall be provided with a dripproof protection enclosure, with one or more doors providing access to internal component for adjustment, test, and replacement.

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(d) Power supply shall be equipped with two mounting attachment points on lower front edge and two near upper back edge. The center of gravity of the assembled power supply shall lie in the plane defined by these four attachment points. Mounting surfaces at these attachment points shall be horizontal so that sound isolation mounts can be easily attached. The equipment structure shall be self-contained and sufficiently rigid to permit the 4-point mounting described above without need for external stiffeners. The power supply shall also be provided with deck mounting feet or runners so that it can be deck mounted without support or sway bracing from the upper rear attachment points.

(e) Provisions for entrance of external wiring for input, output, and remote control (see 3.3.2) through the bottom of the enclosure shall be made.

(f) The complete power supply shall not weigh more than 1250 pounds.

(g) Lifting eyes or lugs shall be provided and shall be suitable for handling the complete power supply and major subassemblies.

3.3.2 Miscellaneous control devices. Miscellaneous control devices shall be provided as follows:

(a) Input magnetic contactor conforming to MIL-C-2212.
(b) On-off pushbuttons or enabling silicon controlled rectifiers for operation of contactor, as specified (see 6.2).
(c) Terminal block and fuse and fuseholder for connection of remote on-off pushbutton station for control of contactor.
(d) Time delay relay with time-on-dropout contacts arranged to reenergize input contactor if input power is interrupted and then restored before the relay has "timed out". The delay time shall be adjustable from 0.2 to 30 seconds. "Warning" plate stating that "stop" pushbutton (see 3.3.2(b)) must remain depressed for a time greater than the time setting of time delay relay shall be provided.
(e) Screwdriver-adjusted rheostat (with locking device) for adjustment of output voltage over specified range. Rheostat and screwdriver adjustment shall be internal to the power supply enclosure.
(f) Terminal block and fuse and fuseholder for connection of remote rheostat identical to that of (e) to permit remote adjustment of output voltage. These terminals and associated circuits shall provide for connection of a remotely located transfer switch to be used to select and make active either the adjustment rheostat contained within the power supply or that located remotely from the power supply.
(g) "Power on" indicator light with white lens, illuminated when power supply is energized.
(h) Ammeter (0-500 A) and voltmeter (0-50 V) (see 3.3.2.1).

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(i) An ammeter "press to read" switch shall be provided. The switch shall consist of a momentary contact toggle switch in accordance with MIL-S-3950. Normal position of switch contacts shall disconnect the ammeter from the load circuit to require operation of the switch to read the load current. The switch shall be spring return to normal position.

3.3.2.1 Digital meters. Digital instruments shall provide a 4-digit readout. The display digits shall be no smaller than 5/32 inch high and 1/8 inch wide. A reset device shall be provided for the digital display to enable resetting of the indicator in place. Means shall be provided for calibration of the meter. Each digital instrument shall be a self-contained assembly mounted on the front panel of the power supply. The maximum overall dimensions of the assembly shall not exceed 4 inches in length and 2.76 inches in height. The bezel of the indicator shall be constructed of metal or plastic material. The metal shall be corrosion-resistant, or shall be treated for corrosion resistance in accordance with MIL-E-917. The plastic material shall be type MMI, MAI, MSG, MSI, or SDG-F in accordance with MIL-M-14. The weight of the assembly shall not exceed 3 pounds. Power required to operate the digital instrument assembly shall be supplied from within the power supply and shall consist of a 115 V, 60 hertz (Hz), single phase source in accordance with Section 300 of DOD-STD-1399. System calibration shall not be affected by voltage excursion from 103.5 V to 126.5 V.

3.3.3 Maintainability.

3.3.3.1 Modular assemblies. The power supply shall be designed so that corrective maintenance can be performed by replacement of modular assemblies. The functional electronic circuitry of the power supply shall be implemented by means of Standard Electronic Modules having the design and physical dimensions specified in MIL-STD-1389. Modules shall be class I in accordance with MIL-M-28787 for submarine application. Modules introduced into the power supply shall be selected in accordance with specific detail requirements as defined by MIL-M-28787. Equipment design shall conform to the design-for-maintainability requirements of MIL-STD-470 and NAVSEA 0967-LP-312-8010.

3.3.3.2 Built-in test assembly. The power supply shall incorporate a built-in test assembly to monitor ripple content of output voltage at the terminals of the power supply. The built-in test assembly shall consist of a detector which monitors ripple voltage continuously. The test assembly shall have the capability to detect total rms voltage ripple and rms value of individual frequency components of voltage ripple at 6th, 12th, and 18th harmonics of power frequency. The test assembly output signal shall be brought out at a panel mounted instrument which is calibrated to read rms voltage and provides a continuous numerical readout of ripple voltage. A selector switch shall be provided for selection of voltage ripple amplitude to be displayed on the numerical readout instrument. Selector switch positions shall be as follows:

(a) Rms total ripple voltage.
(b) Rms 6th harmonic voltage.
(c) Rms 12th harmonic voltage.
(d) Rms 18th harmonic voltage.

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The test selector switch shall be located adjacent to the numerical readout instrument. Selector switch and instrument for readout of voltage ripple shall be mounted on the front panel of the power supply. Test points in accordance with MIL-P-15736 shall also be furnished for measurement of rms voltage ripple and individual frequency components of ripple by means of portable test equipment external to the power supply.

3.3.4 Reliability. Mean-time-between-failure (MTBF) of the power supply shall be not less than 10,000 hours.

3.3.5 Shock. Equipment shall meet the high impact shock requirements for grade A, deck mounted class I medium weight equipment, as determined by type A test of MIL-S-901.

3.3.6 Structureborne noise. Structureborne noise generated by the equipment shall not exceed the limits for type 3 equipment in accordance with MIL-STD-740, when the power supply is operated at any load between no load and rated full load A d.c.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-P-15736. The power supply shall be subjected to quality conformance with table II, as inspection of special output performance of the equipment.

TABLE II. Special output performance inspection.1/

Description	Selective	Periodic	Qualification	Requirement paragraph	Inspection paragraph
Input power interruption	X	X	X	3.2.1.1	4.2.1
Harmonic distortion	-	X	X	3.2.1.2	4.2.2
Burn-in	X	-	X	3.2.2.1	4.2.6
Overcurrent protection	X	X	X	3.2.2.4	4.2.3
Output voltage ripple content	-	X	X	3.2.2.3	4.2.4
Overvoltage limit	-	X	X	3.2.2.1(e)	4.2.5

1/ The power supply shall be operated at room temperature from a 440 V, 3-phase, 60 Hz source. Output voltage shall be regulated at the d.c. output terminals of the power supply by means of local sensing.

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4.2 Output performance inspection.

4.2.1 Input power interruption. The power supply shall be energized through a power supply interrupt device consisting of a power transfer switch, type ABT-A1, in accordance with MIL-S-17773 and rated at 100 A. The rectifier power supply shall be subjected to transfer and retransfer between unsynchronized a.c. voltage lines at each combination of output voltage setting and load specified in table III, test conditions No. 1 through 12 respectively. Ten cycles of transfer and retransfer of the d.c. power supply shall be performed at each test condition stated in table III. Each cycle is defined as follows:

(a) Transfer. The power supply shall initially be supplied by the normal a.c. voltage line. After steady state operation has been established, the power supply shall be transferred to the alternate a.c. voltage line.

(b) Retransfer. After 1 minute of steady state operation on the alternate a.c. voltage line, the power supply shall be retransferred from alternate to normal a.c. voltage line. The power supply shall continue to operate for 5 minutes on the normal a.c. voltage line before the cycle is repeated.

TABLE III. Input power interruption.

Test condition No.	Period of power interruption	Load 1/	Output voltage setting
1		No load	32 V d.c.
2		No load	34 V d.c.
3	Off-time for type	No load	28 V d.c.
4	ABT-A1 (100 A)	No load	30 V d.c.
5	is 20 ms	240 A	34 V d.c.
6	50 ms.	240 A	26 V d.c.
7	Transfer time is	240 A	32 V d.c.
8	1/2 second for trans-	240 A	28 V d.c.
9	fer and retransfer	240 A	34 V d.c.
10	of d.c. power	480 A	28 V d.c.
11	supply	480 A	26 V d.c.
12		480 A	32 V d.c.

4.2.1.1 Output terminal voltage of the power supply shall be regulated by means of local sensing. Input voltage, power supply output terminal voltage, and output current shall be recorded by means of oscillograms having sufficient scale of amplitude and time to provide good readability. The recorded data shall demonstrate that the d.c. output voltage of the power supply equipment conforms to 3.2.2.1(b) during steady state operation following transfer and retransfer of the power supply between a.c. voltage lines.

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4.2.2 Harmonic distortion. Power supply equipment shall be tested to determine compliance with the limitations in allowable magnitudes of harmonic line current specified in 3.2.1.2.

4.2.2.1 Input voltage distortion. The a.c. input line which feeds the power supply shall be monitored for distortion. The power source shall not have a harmonic voltage content at any frequency in percent rms which is greater than 10 percent of the allowable percentage harmonic current at the frequency, as stated in 3.2.1.2, when the a.c. line supplies balanced full load line current to a three phase 20 kW resistive load.

4.2.2.2 Input current measurement. Power supply equipment shall be tested to determine compliance with the limitations in allowable magnitudes of harmonic line current specified in 3.2.1.2. The power supply under test shall be connected to a 440 V, 3-phase, 60 Hz power source. The power supply shall be operated at no load, 50 percent load, and rated full load, respectively, and shall furnish d.c. output voltage at a setting of 32 V d.c. The following minimum data shall be recorded for each output load: d.c. output voltage, output load current, a.c. input line voltage, fundamental frequency input line current, and harmonic frequency components of input current between the second harmonic and 20 kHz. Harmonic line currents shall be plotted against frequency. Fundamental and harmonic currents shall be measured by determining the voltage developed across a non-inductive line resistor in each phase. The resistance value of the line resistor shall be such that the potential drop at the fundamental frequency will be approximately 1 percent of line voltage. The accuracy of measurement of harmonic current shall be 5 percent. A Hewlett Packard Model 3580 spectrum analyzer, or equivalent, shall be used for voltage measurement. This test shall demonstrate that the power supply conforms to 3.2.1.2 with respect to harmonic distortion and 3.2.2.1 with respect to d.c. output voltage.

4.2.3 Overcurrent protection. The power supply shall be connected to output load terminals through the line resistance shown on figure 1. Load cable between power supply and load shall be sized at 225 A. Circuit breaker AQB-A250, with trip element rated at 225 A d.c, shall be wired in series with resistor load bank as shown on figure 1 and shall be closed. Switch S1 shall initially be open. Output voltage at the terminals of the power supply shall be 29 V d.c. The power supply shall be operated with local sensing to regulate d.c. voltage at the output terminals of the power supply equipment. The power supply shall supply 150 A to the load, which consists of load resistor R1. Output voltage line drop between power supply and load shall be 1 V at 150 A d.c. Voltage developed across the load shall be 28 V d.c, when resistor R1 carries 150 A. When the test circuit of figure 1 has been established, switch S1 shall be closed. This switch closure, which places resistor R2 in parallel with resistor R1, shall cause the power supply to furnish short circuit current of 720 A d.c. to the load for purposes of tripping the AQB-A250 circuit breaker. The instantaneous trip element of the AQB-A250 circuit breaker shall be adjusted at maximum setting. Fault current which flows through the circuit breaker shall be recorded by an oscillograph. The method by which the fault is cleared shall be recorded. The d.c. output voltage of the power supply shall be recorded after the fault has been cleared. The manner in which the fault is cleared and subsequent operation of the power supply shall demonstrate conformance to 3.2.2.4.

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4.2.4 Output voltage ripple content. In addition to the test conditions for measurement of ripple specified in MIL-P-15736, rms ripple content of output voltage of the power supply shall be examined at 25 percent and 75 percent load respectively. This test shall demonstrate compliance with 3.2.2.3.

4.2.5 Overvoltage protection. Analysis of power supply design shall be performed in order to confirm that component failures, occurring singly or in combination, will not drive the power supply output voltage to a value greater than 35 V d.c. (see 3.2.2.1(e)).

4.2.6 Burn-in. The power supply shall be subjected to a burn-in test of 96 hours duration. The burn-in test shall be performed as a selective and qualification test in accordance with MIL-P-15736. Each power supply under test shall be operated for a period of 96 hours under the following conditions of voltage and load:

(a)	Output voltage and load:	28 V d.c. at 480 A d.c, initial loading at power supply terminals.
(b)	Temperature:	Room ambient
(c)	Input line voltage:	440 V (nominal) with plus or minus 3 percent unbalanced between line-to-line voltage.

The tests specified in table IV shall be conducted during the 96 hour operational burn-in period at intervals as indicated:

TABLE IV. Burn-in tests.

Test	Test paragraph	Intervals
Input power interruption (Test condition No. 8)	4.2.1	24 hours
Loading cycling	4.2.7	12 hours
Random frequency vibration	See MIL-STD-781	

Random frequency vibration test shall consist of motion input of 6 g rms acceleration in accordance with MIL-STD-781. Duration of vibration shall be 10 minutes within each hour of equipment operating time. Output d.c. voltage at the output terminals of the power supply shall be monitored during the burn-in test in order to determine compliance with 3.2.2.1. Any component failures occurring during the burn-in of the power supply shall be reported, with explanation of cause of failure and recommendations for corrective action to prevent recurrence of the same failure.

4.2.7 Load cycling. The power supply shall be initially loaded at 480 A d.c. and shall be adjusted at 28 V d.c. regulated set point at the output terminals of the power supply. The load shall be cycled 24 times between 480 A d.c. load and 240 A d.c. load. Each cycle shall consist of operation at 480 A load for an interval between 0.8 second and 1.2 seconds, followed by operation at 240 A load for an interval of 10 seconds to complete the cycle. Voltage deviation and transient recovery time shall be recorded by means of oscillograms of output voltage and load current having sufficient scale of amplitude and time to provide good readability. The recorded data shall demonstrate that the power supply equipment conforms to 3.2.2.1(c).

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5. PACKAGING

5.1 Preparation for delivery shall be in accordance with MIL-P-15736.

6. NOTES

6.1 The notes specified in MIL-P-15736 are applicable to this specification.

6.2 Ordering data. In addition to the ordering data requirements of MIL-P-15736, the acquisition documents should specify the following:

(a) Description of circuit breakers used in the d.c. feeders which distribute weapon system 28 V d.c. voltage to the launcher control panels on board submarines where power supply will be installed.
(b) Whether on-off pushbuttons or enabling silicon controlled rectifiers for operation of contactor is required (see 3.3.2).

6.3 Input power interruption. Missile fire control power supplies are a vital load on the submarine electric plant. In order to obtain maximum reliability and continuity of a.c. line voltage during battle or major casualties, all vital electrical loads are provided with limited break ship's service a.c. power. The guided missile fire control power supplies are supplied by two independent ship's service turbine generators (normal and alternate). Upon loss of the normal ship's service generator, vital loads are automatically transferred to the alternate ship's service generator (which is not synchronized with the normal ship's service generator) by means of automatic bus transfer (ABT) switches. After the normal ship's service generator has been restored, the ABT operates automatically to retransfer vital loads from alternate to normal ship's service generator. A time delay is introduced in the transfer operation of all bus transfer switches to avoid excessive current transients when switching from one generator to the other. A time interval of 1/2 second is typically required to transfer the power supply from normal to alternate ship's service generator. (A similar time interval is required to retransfer the power supply from alternate to normal ship's service generator.) The 1/2 second transfer time of the automatic bus transfer includes an off-time between 20 ms and 50 ms. The fire control power supply therefore experiences interruption of a.c. line voltage during ABT. D.c. output voltage at the utilization load equipment may not fall to zero during the ABT function on the a.c. input line to the power supply. The power supply is required to be designed so that when the 440 V, 3-phase, 60 Hz line is interrupted and restored, no damage will result to any component (see 3.2.1.1). After restoration of a.c. power line voltage, the fire control rectifier power supply is required to perform its intended function (see 3.2.1.1). Interruption of a.c. line voltage may occur during training exercises of electrical power system tests. Transfer of load from normal to alternate ship's service generator, and retransfer of load from alternate to normal ship's service generator, can be initiated manually by operating the same bus transfer switch in manual mode with automatic function disabled.

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6.4 Overcurrent protection. When installed on board the submarine, the power supply will be required to furnish ship's service d.c. voltage through power distribution panels to weapons systems consisting of launcher control panels, missile fire control switchboards, and missile test instrumentation. In the event of an overload or short circuit fault on a local feeder at a power distribution panel, the power supply is expected to furnish sufficient fault current to force local clearing of the fault on the feeder. The launcher control panels constitute the largest concentration of load on the d.c. bus. The equipment specified herein is required to supply sufficient fault current to trip the local breaker in the event that a short circuit fault occurs at Launcher Control Panel No. 1 or No. 2 (see 3.2.2.4). These panels are each interrupted by a circuit breaker type ALB-10 having a 150 A trip element. The breakers are located in the power distribution panel local to the launcher control panels. Type ALB-10 circuit breakers furnished in accordance with MIL-C-17588C(SH) are obsolete and are scheduled to be replaced by type AQB circuit breakers in accordance with MIL-C-17361. During the transition between type ALB and type AQB circuit breakers, some ships may continue to operate with ALB circuit breakers after installation of power supplies of new manufacture rated at 480 A d.c. It is therefore suggested that the power supply be tested for capability to force the tripping of ALB-10 circuit breakers to clear a short circuit at the load. For purposes of this test, the AQB-A250 circuit breaker of figure 1 should be replaced by a Type ALB-10 circuit breaker with 200 A trip element and resistor R2 of figure 1 should be 0.0487357 ohms.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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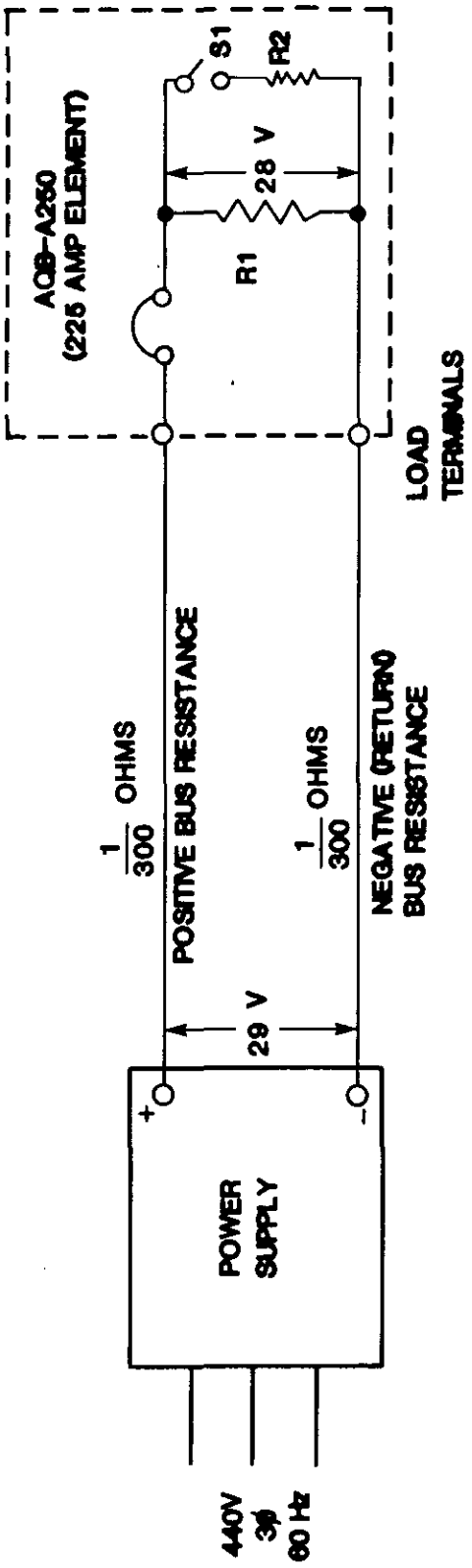


FIGURE 1. Test circuit for overcurrent protection.

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